LMS-Virtual University of | (1) Badar Toor | Badar
Secure | https://lms2.vu.edu.pk/WebTop/Home.aspx?id=b8839d45-0cd0-4950-b1db-900415b47c22

Index Module 133 To 183

133	Criteria for T_1 space	154	Cofinite topology and compactness	175	Examples:Connected component
134	Subspace of T_1 space	155	Compactness and open collection	176	Connected component is closed
135	Properties of T_1 space	156	Closed subspace of compact space	177	Connected component is not open
136	Properties of T-1 space: II	157	Compact subspace of Hausdorff space	178	Locally connected spaces
137	T-2/Hausdorff spaces	158	Compact space under continuous map	179	Examples: Locally connected spaces
138	Examples: hausdorff spaces	159	Compactness and Homomorphism	180	Connectedness and local Connectedness
139	Subspace of Hausdorff spaces	160	Application: Compactness and Homomorphism	181	Path connected spaces
140	Metric spaces are Hausdorff spaces	161	Finite union of Compact spaces	182	Facts: Path connected spaces 1
141	T1 does not imply T2	162	Compact Hausdorff space is normal	183	Facts: Path connected spaces 2 Current Lesson
142	Unique Limit point	163	Separated sets		
143	Unique limit point theorem	164	Examples:Separated sets		
144	Regular Spaces	165	Connected spaces		
145	Example: Regular Spaces	166	Examples: Connected spaces		
146	Normal Space	167	Connected spaces and open closed sets		
147	Examples: Normal spaces	168	Connected sets and its closure		
148	Metric spaces are normal	169	Image of connected space		
149	T_i properties: An overview	170	Application: Image of connected set		
150	Urysohn's Lemma	171	Connected subsets of real line		
151	Compact Spaces	172	Connectedness and fixed point theorem		
152	Examples: Compact space	173	Union of connected subsets		
153	Open intervals in R	174	Connected component		
154	Cofinite topology and compactness				

MTH 634

Module 133 To 183

for Final Term

Created by Badar Toor

MSc Math Bahawalnagar

Activate Windows
Go to PC settings to activate Windows.

fb.com/badartoor

MTH 634 Module ... | LMS-Virtual Unive... | ENG | 11:02 PM 9/5/2018

Criteria for T_1 Space



We have discussed T_1 space and its examples.
Now we will state and prove a very simple criteria to check T_1 property in some space.

Activate Windows
Go to PC settings to activate Windows.

Criteria for T_1 Space

Theorem:

A topological space $(X, \mathcal{T})$ is T_1 iff every singleton subset $\{x\} \subset X$ is closed.

Activate Windows
Go to PC settings to activate Windows.

Subspace of T_1 Space

Theorem:
Every subspace of a T_1 space is also a T_1 space.

Activate Windows
Go to PC settings to activate Windows.

Properties of T_1 Spaces: I



Let us discuss some properties of T_1 spaces. Some of these properties can also be used as a criterion of T_1 property.

Property I: Every finite subset of T_1 space is closed.

Activate Windows
Go to PC settings to activate Windows.

Properties of T_1 Spaces: I

Property II:

Let X be a finite set. Then (X, T) is a T_1 space if and only if (X, T) is discrete space.

Proof: Let X is finite and (X, τ) is T_1 space

See Video Lectures for Proof

Activate Windows
Go to PC settings to activate Windows.

Properties of T_1 Spaces: II

Property III:
Let A be a finite subset of a T_1 space X then A' is empty.
Proof:
Let A is finite subset of X, say A has n elements, i.e.

$$\{a_1, \ldots, a_n\}$$

Since A is finite and X is a T_1 space and we know that every finite subset of a T_1 space is closed so

$$A' \subset A$$

Now let us check one by one the points of A as limits points of A.

Activate Windows
Go to PC settings to activate Windows.

Properties of T_1 Spaces: II

Let's start with a_1.
Now note that

$$U_{a_1} = \{a_2, \ldots, a_n\}^c$$

is an open set containing a_1 and

$$A \cap (U_{a_1} \setminus \{a_1\}) = \emptyset$$

So a_1 is not a limit point of A.
Similarly, no other point of A is a limit point of A. So

$$A' = \emptyset$$

Activate Windows
Go to PC settings to activate Windows.

T_2/Hausdorff Spaces



Felix Hausdorff
(1868 - 1942)

Def:
A topological space $(X, \mathcal{T})$ is said to be "T_2 Space" iff for every distinct pair of elements $x, y \in X$ there exist open subsets U_x, U_y of X containing x, y respectively such that $U_x \cap U_y = \emptyset$.

T_2 spaces are also called Hausdorff Spaces.

Activate Windows
Go to PC settings to activate Windows.

Examples: Hausdorff Spaces



In this topic we will discuss some examples of Hausdorff spaces.

Example 1:

Every discrete space is Hausdorff.

Example 2:

A nonempty set $X \neq \{x\}$ with indiscrete topology is not Hausdorff.

Activate Windows
Go to PC settings to activate Windows.

Examples: Hausdorff Spaces

Example 3: $\mathbb{R}$ with lower limit topology is Hausdorff.

$[a, b)$ (a, b)

$x, y \in \mathbb{R}$ $x < y$

$x \in [x, c), [y, d) \ni y$ $x < c < y < d$

$U_x \cap U_y = \emptyset$



Activate Windows
Go to PC settings to activate Windows.

Examples: Hausdorff Spaces

Example 4: $\mathbb{R}$ with topology generated

$$\{(a, \infty) | a \in \mathbb{R}\}$$

is not Hausdorff.

$3, 6 \in \mathbb{R}$

$6 \in (5, \infty)$

$3 \in (a, \infty) \not\ni 6$ $\quad a < 3$

Activate Windows
Go to PC settings to activate Windows.

Examples: Hausdorff Spaces



Example 5: An infinite set with cofinite topology is not Hausdorff space.

MTH 634

Module 133 To 183
for Final Term

Created by Badar Toor

MSc Math Bahawalnagar

fb.com/badartoor

Activate Windows
Go to PC settings to activate Windows.

Topology



Subspace of Hausdorff Space

MTH634_Topic139.mp4

Activate Windows
Go to PC settings to activate Windows.

Subspace of Hausdorff Space

Theorem:
Let (X, T) is a Hausdorff space and $A \subset X$ then (A, T_A) is Hausdorff.

See Video Lectures for Proof

Proof: Let $A \subset X$

Given (X, T) is Hausdorff.

$\Rightarrow \forall \; x, y \in X \quad x \neq y$

$\exists \; U_x, U_y \in T \Rightarrow U_x \cap U_y = \phi$

$x \in U_x$, $y \in U_y$

$a, b \in A \subset X \quad \exists \; U_a, U_b \in T$ s.t

$U_a \cap U$

Activate Windows
Go to PC settings to activate Windows.

Metric Spaces are Hausdorff



Recall the topic where we discussed some examples of "Non Metrizable Spaces". We discussed there that an infinite set X with cofinite topology is not metrizable. It is because of non-Hausdorffness of X with respect to cofinite topology. In this topic we will discuss an important property of metric spaces called "Hausdorffness".

Activate Windows
Go to PC settings to activate Windows.

Metric Spaces are Hausdorff

Theorem:
Every metric space is Hausdorff.

Proof

See Video Lectures for Proof

MTH 634

Module 133 To 183
for Final Term

Created by Badar Toor

MSc Math Bahawalnagar

fb.com/badartoor

Activate Windows
Go to PC settings to activate Windows.

$T_1 \not\Longleftrightarrow T_2$



T_2 property is a stronger property than T_1 property.

i.e.

Every T_2 space is T_1 but not every T_1 space is T_2.

Activate Windows
Go to PC settings to activate Windows.

$T_1 \not\Longleftrightarrow T_2$

Example: An infinite set X with cofinite topology is T_1 space but not T_2 space.
Justification: Given X is infinite. Now with respect to cofinite topology on X, a subset D of X is open in X iff

$$D = \emptyset \quad \text{or} \quad D^c = \text{finite}$$

So for each pair of distinct points x, y in X one may consider

$$U_x = \{y\}^c \quad \text{and} \quad U_y = \{x\}^c$$

and it is clear that

$$x \in U_x \not\ni y \quad \text{and} \quad y \in U_y \not\ni x$$

Activate Windows
Go to PC settings to activate Windows.

$T_1 \not\Longrightarrow T_2$

So $(X, \mathcal{T}_{cof})$ is T_1.
Now we will show that $(X, \mathcal{T}_{cof})$ is not T_2.
Lets start contrarily i.e. with the assumption $(X, \mathcal{T}_{cof})$ is T_2. So for any distinct pair of points x, y in X, there exist U_x, U_y (open subsets of X containing x and y resp.), s.t.

$$U_x \cap U_y = \emptyset$$

Now $(U_x)^c$ and $(U_y)^c$ are finite.Now

$$(U_x \cap U_y)^c = (U_x)^c \cup (U_y)^c$$

is finite. But $(U_x \cap U_y)^c = X$ that is infinite.
Contradiction. So $(X, \mathcal{T}_{cof})$ is not T_2.

Activate Windows
Go to PC settings to activate Windows.

Unique Limit Point



Recall the definition of "Convergent Sequence". Let $(X, \mathcal{T})$ be a topological space. A sequence of points $(x_i)_{i \in I}$ in X is said to be converges to a point $x \in X$ if for every open subset U_x, containing x, in X there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n \in U_x$.

Activate Windows
Go to PC settings to activate Windows.

Unique Limit Point



Now if we consider convergent sequences in $\mathbb{R}^n$ with usual topology then our experience for such sequences is uniqueness of the limit points.
But if we critically analyze the definition of convergent sequences, we find no such condition on the uniqueness of the limit point.

Activate Windows
Go to PC settings to activate Windows.

Unique Limit Point

Example:
Consider $X = \{a, b, c\}$ with

$$\mathcal{T} = \{\emptyset, \{b\}, \{a, b\}, \{b, c\}, X\}$$

The constant sequence (b) converges to a, to b, and to c.

Activate Windows
Go to PC settings to activate Windows.

Unique Limit Point

Topological spaces in which sequences converge to unique limit points are of more importance than the topological spaces in which limit are not unique. And the reason is, very limited results can be stated/proved in spaces with multiple limit points. So one should impose some extra condition/property on the space to avoid such multiplicity. And this condition/property brings the class of spaces under consideration closer to which one's geometric intuition applies.

The property that ensure the uniqueness of limit point is the Hausdorff property.

Activate Windows
Go to PC settings to activate Windows.

Theorem: Unique Limit Point



Theorem:
Let X be a Hausdorff space then every convergent sequence of points has a unique limit point.
i.e.
for (x_i) in X if

$$a, b \in X$$

are the limit points of (x_i) then

$$a = b.$$

Activate Windows
Go to PC settings to activate Windows.

Theorem: Unique Limit Point

Proof:

Consider a convergent sequence (x_i) in a Hausdorff space X. Let (x_i) converges to $a, b \in X$ and

$$a \neq b$$

Since X is Hausdorff. So for $a, b \in X$ $\exists$ a pair of open subsets, U_a and V_b, of X, such that

$$U_a \cap V_b = \emptyset$$

Now if a is the limit of (x_i) then

$\exists\ N \in \mathbb{N}$ such that $\forall\ n \geq N \Rightarrow x_n \in U_a$

Activate Windows
Go to PC settings to activate Windows.

Theorem: Unique Limit Point

And if b is the limit of (x_i) then

$$\exists \ M \in \mathbb{N} \text{ such that } \forall \ n \geq M \ \Rightarrow \ x_n \in V_b$$

So it means that $\forall \ n \geq \max\{M, N\}$ we have

$$x_n \in U_a \cap V_b$$

Which is the contradiction of the fact

$$U_a \cap V_b = \emptyset$$

So $a = b$, i.e. in a Hausdorff space then every convergent sequence of points has a unique limit point.

Activate Windows
Go to PC settings to activate Windows.

Regular Spaces



Def:

A topological space (X, T) is said to be "regular" iff for every closed subset F of X and for every $x \in F^c$ there exist open subsets U_F and U_x containing F and x respectively such that

$$U_F \cap U_x = \emptyset.$$

Activate Windows
Go to PC settings to activate Windows.

MTH634_Topic144.mp4

Regular Spaces



T_3 Space:

A topological space $(X, \mathcal{T})$ is said to be "T_3 space" iff it is either

Regular+T_0

or

Regular+T_1

or

Regular+T_2

Activate Windows
Go to PC settings to activate Windows.

Regular Spaces

Remarks:

1. Property of being regular is a topological property.
2. Every subspace of a regular space is a regular space.
3. T_3 property is a topological property.
4. Every subspace of a T_3 space is a T_3 space.

Activate Windows
Go to PC settings to activate Windows.

Topology



Examples: Regular Spaces

MTH634_Topic145.mp4

Activate Windows
Go to PC settings to activate Windows.

Examples: Regular Spaces



In this topic we will discuss some examples of regular spaces.

Example 1:

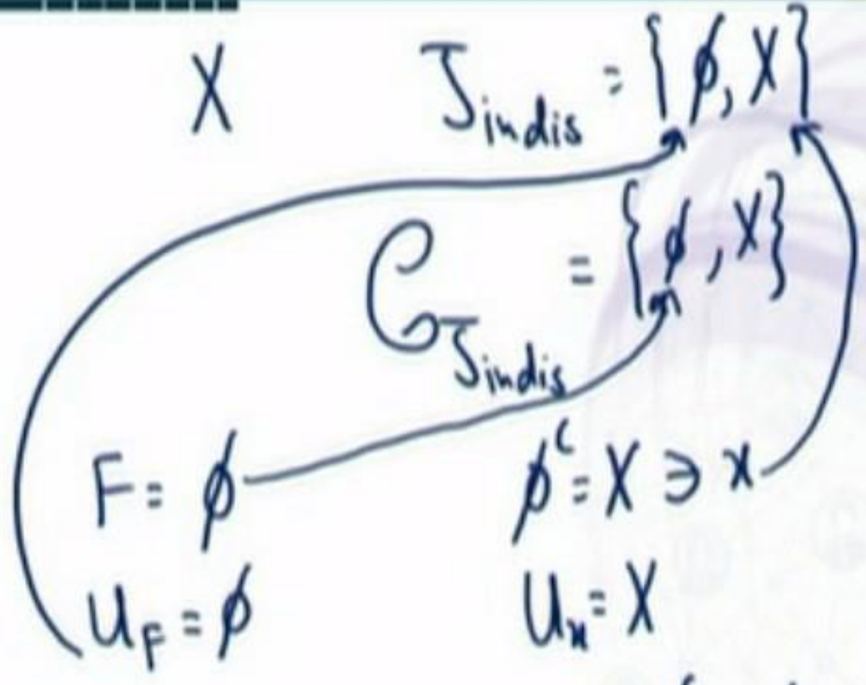
Discrete space is a regular space.

Example 2:

Indiscrete space is a regular space.

Activate Windows
Go to PC settings to activate Windows.

Examples: Regular Spaces



$\emptyset \cap X = \emptyset$

$X \neq \{x\}$

But $(X, \mathcal{T}_{indis})$ is not T_3-Space $\equiv$ Regular + T_2.

$\therefore$ $(X, \mathcal{T}_{indis})$ is not T_2

Activate Windows
Go to PC settings to activate Windows.

Examples: Regular Spaces

Example 3: Consider $X = \{a, b, c\}$ with

$$\mathcal{T} = \{\emptyset, \{a\}, \{b, c\}, X\}.$$

$(X, \mathcal{T})$ is a regular space.

Activate Windows
Go to PC settings to activate Windows.

Examples: Regular Spaces

Example 3: Consider $X = \{a, b, c\}$ with

$$\mathcal{T} = \{\emptyset, \{a\}, \{b, c\}, X\}.$$

$(X, \mathcal{T})$ is a regular space.

MTH 634
Module 133 To 183
for Final Term
Created by Badar Toor
MSc Math Bahawalnagar
fb.com/badartoor

Activate Windows
Go to PC settings to activate Windows.

Examples: Regular Spaces



Example 4:
$\mathbb{R}$ with usual topology is regular. Moreover this space is also T_3 space.

Example 5:
An infinite set with cofinite topology is not regular.
So it is not T_3.

Activate Windows
Go to PC settings to activate Windows.

Examples: Regular Spaces

Example 6:
Recall the K-Topology on real line.
Topology $\mathcal{T}_K$ generated by

$$\mathcal{B}_K = \{(a, b) |\ a, b \in \mathbb{R}\} \cup \{(a, b) \backslash K | a, b \in \mathbb{R}\}$$

where

$$K = \{1, 1/2, 1/3, \cdots\} \subset \mathbb{R}$$

$(\mathbb{R}, \mathcal{T}_K)$ is not a regular space.

Activate Windows
Go to PC settings to activate Windows.

Do you want to restart the playback where left off?
Continue
X

Topology



Normal Spaces

Activate Windows
Go to PC settings to activate Windows.

MTH634_Topic146.mp4

Do you want to restart the playback where left off?
Continue
X

Normal Spaces



Def:

A topological space (X, T) is said to be "normal" iff for every pair of disjoint closed subsets, $F_1, F_2 \subset X$ there exist open subsets U_{F_1} and U_{F_2} containing F_1 and F_2 respectively such that

$$U_{F_1} \cap U_{F_2} = \emptyset.$$

Activate Windows
Go to PC settings to activate Windows.
MTH634_Topic146.mp4

Normal Spaces

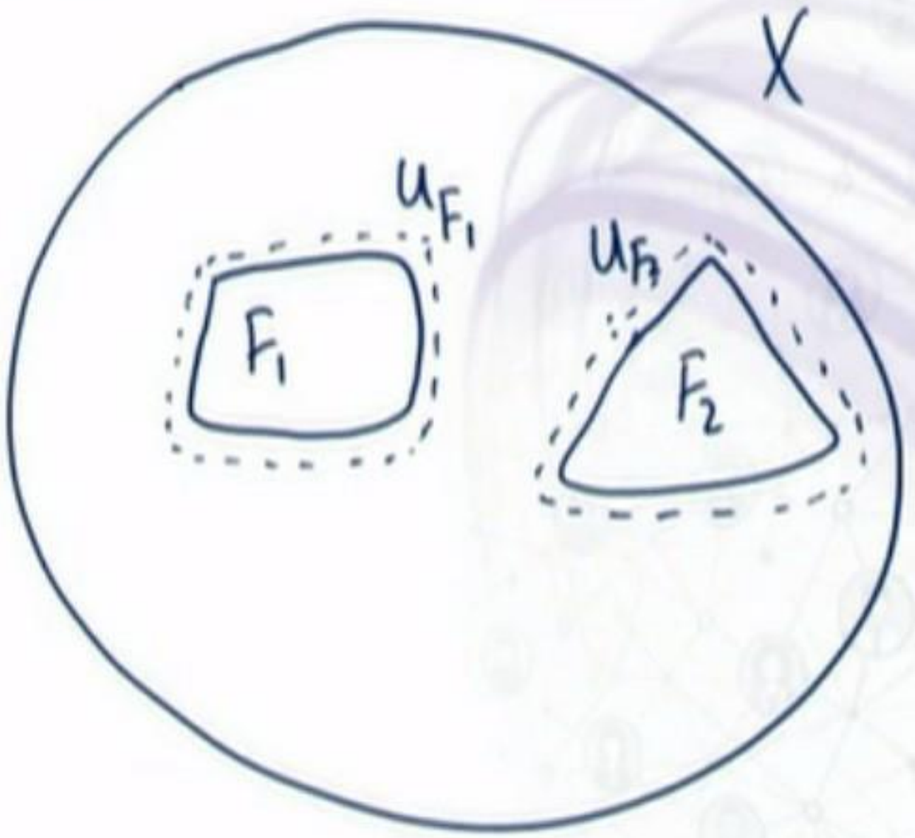


Def:
A topological space (X, T) is said to be "normal" iff for every pair of disjoint closed subsets, $F_1, F_2 \subset X$ there exist open subsets U_{F_1} and U_{F_2} containing F_1 and F_2 respectively such that

$$U_{F_1} \cap U_{F_2} = \emptyset.$$

Activate Windows
Go to PC settings to activate Windows.

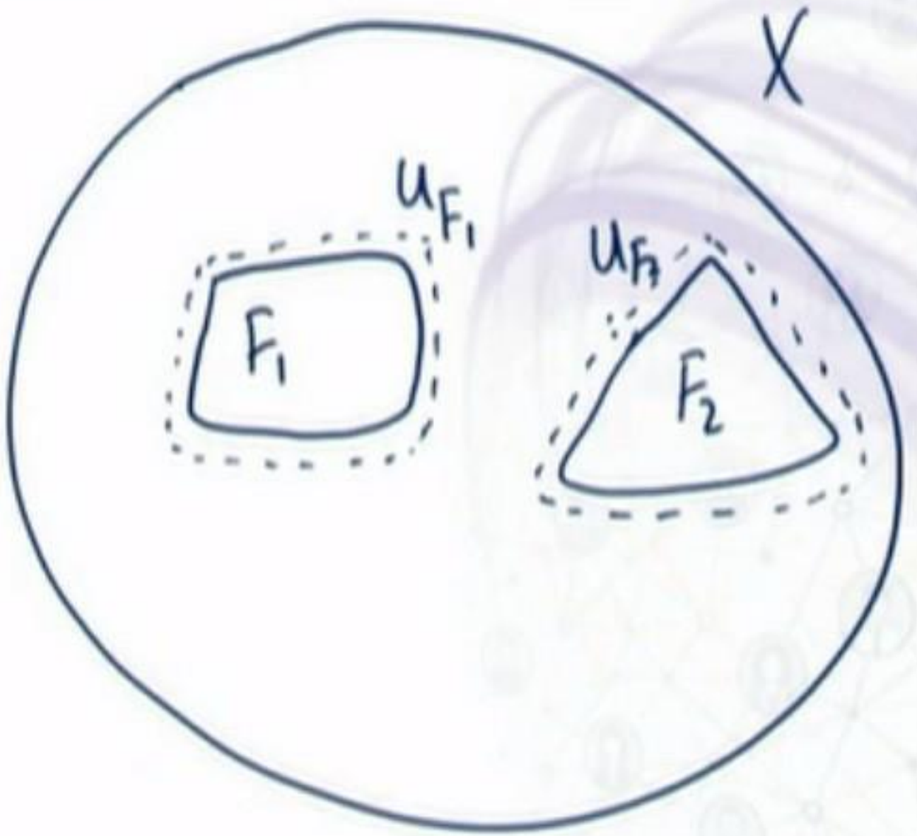
Normal Spaces



$F_1 \cap F_2 = \emptyset$

Activate Windows
Go to PC settings to activate Windows.

Normal Spaces



$F_1 \cap F_2 = \emptyset$

$U_{F_1} \cap U_{F_2} = \emptyset$

Activate Windows
Go to PC settings to activate Windows.

Normal Spaces



T_4 Space:

A topological space $(X, \mathcal{T})$ is said to be "T_4 space" iff it is either
Normal+T_1
or
Normal+T_2

Activate Windows
Go to PC settings to activate Windows.

Normal Spaces

Remarks:

1. Property of being normal is a topological property.
2. Subspace of a normal space needs not be a normal space.
3. T_4 property is a topological property.

Completely Normal Space:
A topological space $(X, \mathcal{T})$ is said to be completely normal space iff every subspace of X with subspace topology is normal.

Activate Windows
Go to PC settings to activate Windows.

Examples: Normal Spaces

Example 1: Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}, X\}.$$

Claim: $(X, \mathcal{T})$ is a normal space.
Explanation:
The closed collection is

$$\mathcal{C}_{\mathcal{T}} = \{\emptyset, \{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}, X\}$$

Activate Windows
Go to PC settings to activate Windows.

Examples: Normal Spaces

Example 1: Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}, X\}.$$

Claim: $(X, \mathcal{T})$ is a normal space. But $(X, \mathcal{T})$ is not T_4

Explanation:

The closed collection is

$$\mathcal{C}_{\mathcal{T}} = \{\emptyset, \{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}, X\}$$

$F_1 = \emptyset$ $F_2 = X$ $U_{F_1} = \emptyset$, $U_{F_2} = X$

$U_{F_1} \cap U_{F_2} = \emptyset$

Activate Windows
Go to PC settings to activate Windows.

Examples: Normal Spaces

Example 2: Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}, X\}.$$

Claim: $(X, \mathcal{T})$ is not T_4 space.

Activate Windows
Go to PC settings to activate Windows.

Examples: Normal Spaces

Example 3: Consider $Y = \{b, c, d\}$ with

$$\mathcal{T}_Y = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, Y\}.$$

Claim: $(Y, \mathcal{T}_Y)$ is not a normal space.

Explanation:

The closed collection is

$$\mathcal{C}_{\mathcal{T}_Y} = \{\emptyset, \{b\}, \{c\}, \{b, c\}, Y\}$$

$F_1 = \{b\}$, $F_2 = \{c\}$

$U_{F_1} \cap U_{F_2} \neq \emptyset$

Activate Windows
Go to PC settings to activate Windows.

Examples: Normal Spaces

Example 4: Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}, X\}.$$

Claim: $(X, \mathcal{T})$ is not a completely normal space.
Explanation:
Consider $Y = \{b, c, d\}$. So the subspace topology on Y with respect to X is

$$\mathcal{T}_Y = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, Y\}.$$

Activate Windows
Go to PC settings to activate Windows.

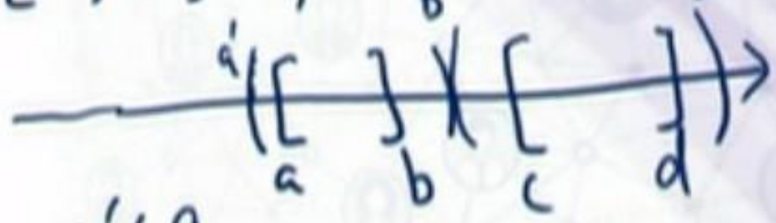
Examples: Normal Spaces

Example 5: $\mathbb{R}$ with usual topology is a T_4 space.

$(\mathbb{R}, \mathcal{T}_u)$ is Normal space.

F_1 F_2

$[a, b] \cap [c, d] = \emptyset$ $\quad b < c$



$U_{F_1} = (a', b')$

$U_{F_2} = (b', d')$

$a' < a$

$b < b' < d$

$d' > d$

$U_{F_1} \cap U_{F_2} = \emptyset$

Activate Windows
Go to PC settings to activate Windows.

Metric Spaces are Normal



In this topic we will discuss another important property of metric spaces, that is called "Normal Property". In fact more than this one metric spaces are T_4 space.

Theorem:
Every metric space is normal.

Activate Windows
Go to PC settings to activate Windows.

Metric Spaces are Normal

Definition:
Let (X, d) be a metric space and A be a nonempty subset of X. Then a continuous function

$$d_A : X \longrightarrow \mathbb{R}$$

defined as

$$d_A(x) = \inf\{d(x, a) | a \in A\},$$

provides the distance between x and A. We denote it as

$$\text{dist}(x, A) = d_A(x)$$

Activate Windows
Go to PC settings to activate Windows.

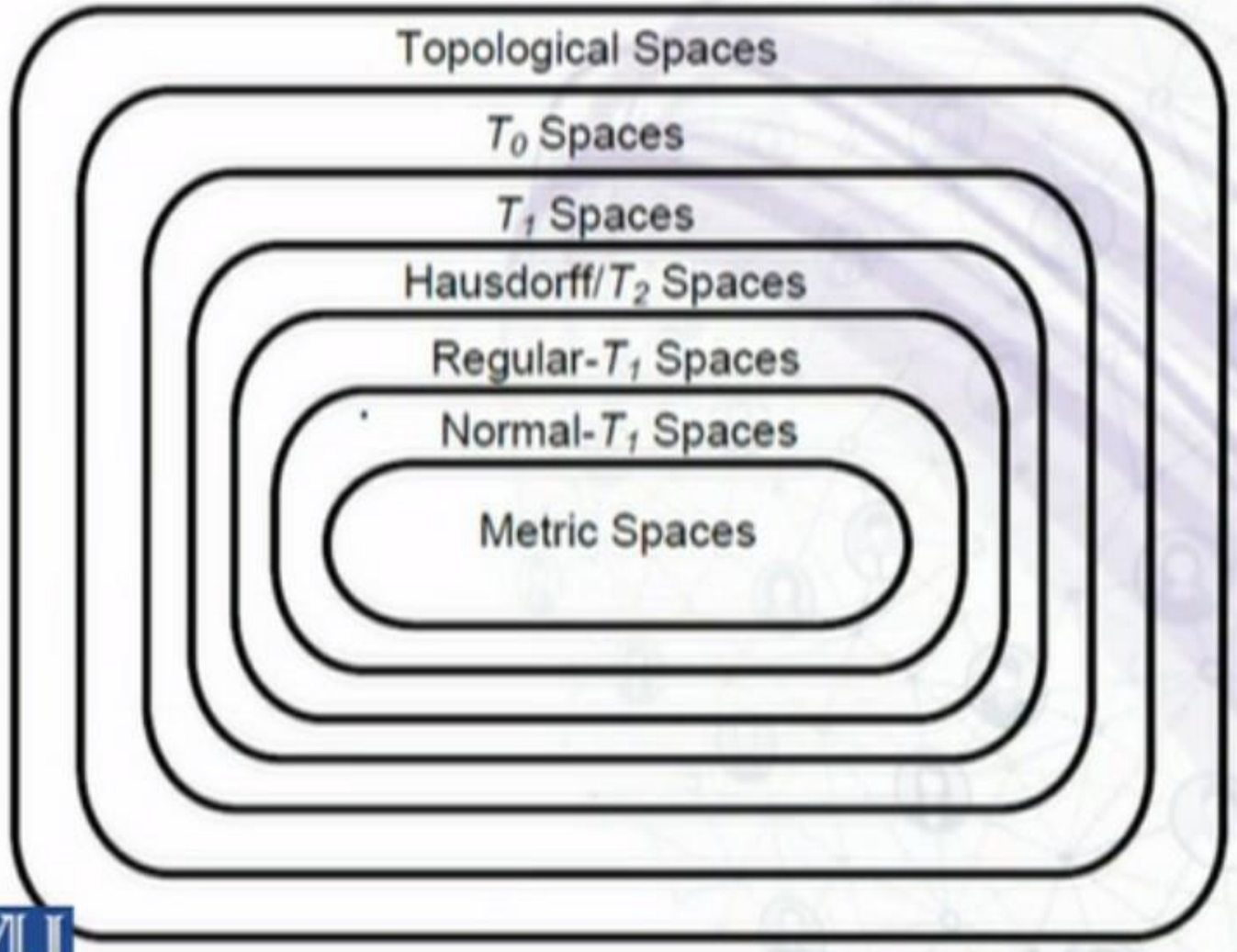
T_i Properties: An Overview



An overview of T_i properties is given in this topic through a diagram. This diagram illustrates the relationship between the T_i spaces we studied in previous topics.

Activate Windows
Go to PC settings to activate Windows.

T_i Properties: An Overview



Activate Windows
Go to PC settings to activate Windows.

Urysohn's Lemma



In this topic, one of the most classical and important results of topology will be discussed. This result is named as "Urysohn's Lemma".
Let us state this lemma.

Activate Windows
Go to PC settings to activate Windows.

Urysohn's Lemma

Urysohn's Lemma: Let X be a normal space; A and B be two disjoint closed subsets of X. Let $[a, b]$ be a closed interval in $\mathbb{R}$. Then there exists a continuous map

$$f : X \longrightarrow [a, b]$$

such that $f(x) = a \ \forall \ x \in A$ and $f(x) = b \ \forall \ x \in B$.

Urysohn's Lemma: (Strong Form)

X is a normal space if and only if for every two disjoint closed subsets A and B of X. There exists a continuous map

$$f : X \longrightarrow [a, b]$$

such that $f(x) = a \ \forall \ x \in A$ and $f(x) = b \ \forall \ x \in B$.

Activate Windows
Go to PC settings to activate Windows.

Urysohn's Lemma

One important application of Urysohn's Lemma is the partial solution of metrization problem. Here we have this solution in the form of "Urysohn's Metrization Theorem".

Urysohn's Metrization Theorem:
Every normal-T_1 space with countable basis is metrizable.

Another version of Urysohn's Metrization Theorem is

Urysohn's Metrization Theorem:
Every regular-T_1 space with countable basis is metrizable.

Activate Windows
Go to PC settings to activate Windows.

Compact Spaces



Recall the definitions of cover, subcover, and open cover.

Cover:

Let $(X, \mathcal{T})$ be a topological space. Let $\mathcal{C} = \{C_i\}$ be a class of subsets of X, such that

$$X = \bigcup_{i \in I} C_i,$$

then $\mathcal{C}$ is called a cover of X.

Activate Windows
Go to PC settings to activate Windows.

Compact Spaces



Subcover:
A subclass $\mathcal{S}$ of a cover $\mathcal{C}$ of X which is also a cover is called a subcover.

Open cover:
A cover $\mathcal{C}$ of X is said to be an open cover if and only if $\mathcal{C} \subset \mathcal{T}$.

Activate Windows
Go to PC settings to activate Windows.

Compact Spaces



Compact Space:
A topological space $(X, \mathcal{T})$ is said to be compact if every open cover $\mathcal{C}$ of X contains a finite subcover $\mathcal{S}$ of X.

Activate Windows
Go to PC settings to activate Windows.

Examples: Compact Spaces



Example 1:
A set X with indiscrete topology $\mathcal{T}$ is compact.
Reason:
Since there is only one open cover of X with respect to indiscrete topology and that is

$$\{X\}$$

which is it self finite.

Activate Windows
Go to PC settings to activate Windows.

Examples: Compact Spaces



Example 2:
A set X with topology $\mathcal{T}$ containing finite number subsets of X is compact.

Reason:
Since every open cover $\mathcal{C}$ of X is a subclass of $\mathcal{T}$ and $\mathcal{T}$ itself finite so $\mathcal{C}$ is finite too and any subcover $\mathcal{S}$ of $\mathcal{C}$ is also finite.

Activate Windows
Go to PC settings to activate Windows.

Examples: Compact Spaces



Example 3:
A finite set X with any topology $\mathcal{T}$ is compact.
Reason:
Since every topology of a finite set X is finite and from "Example 2" X is compact.

Activate Windows
Go to PC settings to activate Windows.

Examples: Compact Spaces

Example 4:

An infinite set X with discrete topology $\mathcal{T}$ is not compact.

Reason:

$\mathcal{T}_{dis} = \mathcal{P}(X)$

$\mathcal{C} = \{\{x\} \mid x \in X\} \cup \{\{x_1, x_2\}\}$ (fin., fin.)

$X \neq \bigcup_{i \in I} C_i$ $\quad C_i = \{x_i\}$

Activate Windows
Go to PC settings to activate Windows.

Open Intervals in $\mathbb{R}$



In this topic we will discuss the compactness of open intervals in $\mathbb{R}$ with respect to the usual topology.

Fact:
An open interval in $\mathbb{R}$ with respect to usual topology is not compact.

An immediate consequence of this fact is

$(\mathbb{R}, \mathcal{T}_u)$ is not compact.

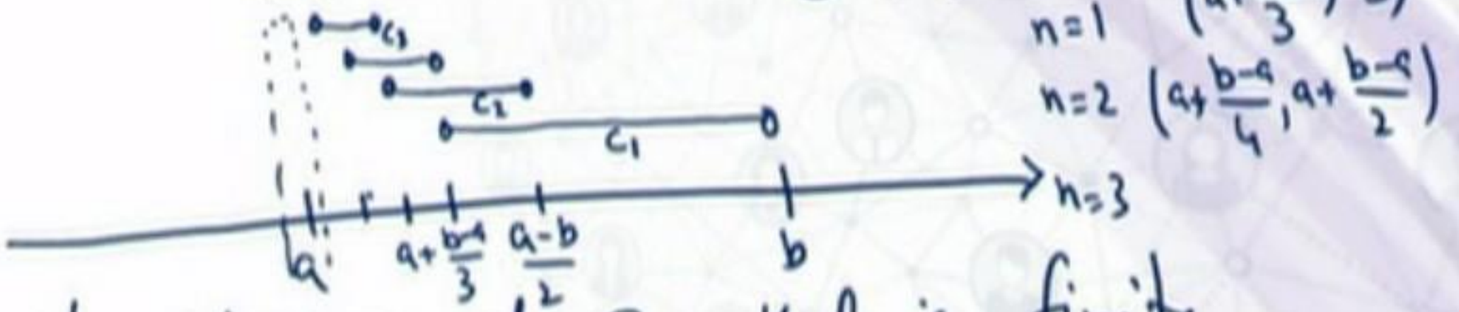
Activate Windows
Go to PC settings to activate Windows.

Open Intervals in $\mathbb{R}$

Proof:

Consider $(\mathbb{R}, \mathcal{T}_u)$ and (a, b) be an open interval in $\mathbb{R}$.
Consider the following covering of (a, b).

$$\mathcal{C} = \left\{ \left(a + \frac{b-a}{n+2}, a + \frac{b-a}{n} \right) \right\} \quad n \in \mathbb{Z}^+$$



$n=1$ $\left(a+\frac{b-a}{3}, b\right)$

$n=2$ $\left(a+\frac{b-a}{4}, a+\frac{b-a}{2}\right)$

So $\nexists$ subcover of $\mathcal{C}$ that is finite

(a, b) is not compact.

Activate Windows
Go to PC settings to activate Windows.

Cofinite Topology and Compactness



In this topic we will study the compactness of cofinite space, a set X with cofinite topology $\mathcal{T}_{cof}$.

Fact:
A set X with cofinite topology $\mathcal{T}_{cof}$ is compact.

Activate Windows
Go to PC settings to activate Windows.

Compactness and Open Collection



The aim of this topic is to show that compactness of some space depends on the collection of open subsets of the space.

Consider $\mathbb{R}$, now

- $(\mathbb{R}, \mathcal{T}_u)$ is not compact.
- $(\mathbb{R}, \mathcal{T}_{indis})$ is compact.
- $(\mathbb{R}, \mathcal{T}_{dis})$ is not compact.
- $(\mathbb{R}, \mathcal{T}_{cof})$ is compact.

Activate Windows
Go to PC settings to activate Windows.

Theorem: Closed Subspace...



Theorem:
Every closed subspace of a compact space is compact.

Proof:
Let $(X, \mathcal{T})$ be a compact space and A be a closed subset of X.
So $(A, \mathcal{T}_A)$ be closed subspace of X, where $\mathcal{T}_A$ is subspace topology on A.
Consider an open cover $\mathcal{D}$ of A, i.e.

Activate Windows
Go to PC settings to activate Windows.

Theorem: Closed Subspace...

$$A = \bigcup_{i \in I} D_i$$

where $D_i \in \mathcal{D} \subset \mathcal{T}_A$.
Now for each D_i there exists $U_i \in \mathcal{T}$ such that $D_i = A \cap U_i$.
So

$$A = \bigcup_{i \in I} D_i = \bigcup_{i \in I} (A \cap U_i)$$

But A is closed in X so A^c is an open subset of X. And we have an open covering of X.

$$X = A^c \cup (\bigcup_{i \in I} U_i)$$

Activate Windows
Go to PC settings to activate Windows.

Theorem: Closed Subspace...

But X is compact. So every open cover of X has a finite subcover, i.e. there exists $J \underset{\text{finite}}{\subset} I$ such that

$$X = A^c \cup (\bigcup_{j \in J} U_j)$$

$$\Longrightarrow$$

$$A \subset \bigcup_{j \in J} U_j$$

$$\Longleftrightarrow$$

$$\underline{A} = A \cap (\bigcup_{j \in J} U_j) = \boxed{\bigcup_{j \in J} D_j}$$

$A \cap U_j = D_j$

Proved.

Activate Windows
Go to PC settings to activate Windows.

Compact Subspace of Hausdorff...



Theorem:
Let $(A, \mathcal{T}_A)$ be compact subspace of a Hausdorff space $(X, \mathcal{T})$.
Then A is closed in X.
i.e.
Every compact subspace of a Hausdorff space is closed.

Activate Windows
Go to PC settings to activate Windows.

Compact Space under Cont. Map



Theorem:
Let $(X, \mathcal{T})$ be a compact space and

$$f : (X, \mathcal{T}) \to (Y, \mathcal{T}')$$

be a continuous map. Then $f(X)$ is compact.
i.e.
Image of compact space under a continuous map is compact.

Activate Windows
Go to PC settings to activate Windows.

Compactness and Homeomorphism



Corollary I:
Let $(X, \mathcal{T})$ be a compact space and $(Y, \mathcal{T}')$ be a Hausdorff space. Then a continuous map

$$f : (X, \mathcal{T}) \to (Y, \mathcal{T}')$$

is a closed map.

Activate Windows
Go to PC settings to activate Windows.

Compactness and Homeomorphism



Corollary II:
Let $(X, \mathcal{T})$ be a compact space and $(Y, \mathcal{T}')$ be a Hausdorff space. Then a bijective continuous map

$$f : (X, \mathcal{T}) \to (Y, \mathcal{T}')$$

is a homeomorphism.

Activate Windows
Go to PC settings to activate Windows.

Application: Compactness and...



Exercise I:
Consider $(\mathbb{R}, \mathcal{T}_u)$. Show that there exists no homeomorphism between an open interval of $\mathbb{R}$ and a closed interval of $\mathbb{R}$.

Exercise II:
Prove or disprove.

$$[0; 1) \approx S^1$$

Activate Windows
Go to PC settings to activate Windows.

Finite Union of Compact Spaces



Theorem:
Let $(X, \mathcal{T})$ be a topological space. And

$$\{(X_i, \mathcal{T}_i) | i \in I\}$$

be a finite family of compact subspaces of X.
Then

$$\bigcup_{i \in I} X_i$$

is compact.

Activate Windows
Go to PC settings to activate Windows.

Compact Hausdorff Space is Normal



Theorem:
Let $(X, \mathcal{T})$ be a compact Hausdorff space. Then $(X, \mathcal{T})$ is normal.
i.e.
For every pair of disjoint closed subsets F_1, F_2 of a compact Hausdorff space X there exists a pair of disjoint open subsets V_{F_1}, V_{F_2} of X containing F_1, F_2 respectively.

Activate Windows
Go to PC settings to activate Windows.

Separated Sets



Def II:
Let A and B be two subsets of a topological space (X, T). Then A and B are said to be separated sets if and only if

$$\overline{A} \cap B = \emptyset$$

and

$$A \cap \overline{B} = \emptyset$$

Activate Windows
Go to PC settings to activate Windows.

Separated Sets



Def III:
Let A and B be two subsets of a topological space (X, T). Then A and B are said to be separated sets if and only if there exist open subsets U_A and U_B of X containing A and B respectively such that

$$A \cap U_B = \emptyset$$

and

$$B \cap U_A = \emptyset$$

Activate Windows
Go to PC settings to activate Windows.

Examples: Separated Sets

Example 1:
Consider $(\mathbb{R}, \mathcal{T}_u)$. Then $A = (0, 1)$ and $B = (4, 9]$ are separated sets.
Reason: (By Definition I)
Since

$$(0, 1) \cap (4, 9] = \emptyset$$

and

$$(0, 1)' \cap (4, 9] = \emptyset$$

and

$$(0, 1) \cap (4, 9]' = \emptyset$$

Activate Windows
Go to PC settings to activate Windows.

Examples: Separated Sets

Let discuss "Example 1" with "Definition III" of separated sets.
There exist $U_A = (-0.5, 2)$ and $U_B = (3, 10)$ two open subsets of $\mathbb{R}$ containing $A = (0, 1)$ and $B = (4, 9]$ respectively such that

$$A \cap U_B = \emptyset$$

and

$$B \cap U_A = \emptyset$$

So A and B are separated sets of $(\mathbb{R}, \mathcal{T}_u)$.

Activate Windows
Go to PC settings to activate Windows.

Examples: Separated Sets

Example 2:
Consider $(\mathbb{R}, \mathcal{T}_u)$. Then $A = (-1, 3]$ and $B = [3, 5)$ are not separated sets.

Reason:
By "Definition I" both sets should be disjoint but this is not the case here, i.e.

$$(-1, 3] \cap [3, 5) = \{3\} \neq \emptyset$$

.

Activate Windows
Go to PC settings to activate Windows.

Examples: Separated Sets

Example 3:
Consider $(\mathbb{R}, \mathcal{T}_u)$. Then $A = (-1, 3)$ and $B = [3, 5)$ are not separated sets.
Reason: (By Definition I)
Both sets are disjoint, i.e.

$$(-1, 3) \cap [3, 5) = \emptyset$$

But at the same time both sets should not contain each others limit points. But here

$$(-1, 3)' \cap [3, 5) = \{3\}$$

Activate Windows
Go to PC settings to activate Windows.

Connected Spaces



Connected Set:
A subset A of a topological space $(X, \mathcal{T})$ is said to be connected iff there exists no pair of nonempty open subsets U and V of X such that

$$A \cap U \quad \text{and} \quad A \cap V$$

are nonempty disjoint sets and

$$A = (A \cap U) \cup (A \cap V)$$

Activate Windows
Go to PC settings to activate Windows.

Connected Spaces



Disconnected Set:
A subset A of a topological space $(X, \mathcal{T})$ is said to be disconnected if it is not connected, i.e.
if there exists a pair of nonempty open subsets U and V of X such that $A \cap U$ and $A \cap V$ are nonempty disjoint sets and

$$A = (A \cap U) \cup (A \cap V).$$

Activate Windows
Go to PC settings to activate Windows.

Connected Spaces



Connected Space:
A topological space $(X, \mathcal{T})$ is said to be connected if it can not be written as union of two nonempty open disjoint subsets U and V of X.

Activate Windows
Go to PC settings to activate Windows.

Connected Spaces



Disconnected Space:
A topological space $(X, \mathcal{T})$ is said to be disconnected if it can be written as union of two nonempty open disjoint subsets U and V of X.

Activate Windows
Go to PC settings to activate Windows.

Connected Spaces



Connected Space: (Def. II)
A topological space (X, T) is said to be connected if

$$X = D_1 \cup D_2$$

such that D_1, D_2 both open and

$$D_1 \cap D_2 = \emptyset$$

implies that either

$$D_1 = \emptyset \quad \text{or} \quad D_2 = \emptyset.$$

Activate Windows
Go to PC settings to activate Windows.

Examples: Connected Spaces

Example 1: Consider $(\mathbb{R}, \mathcal{T}_u)$. Then $A = (0,1) \cup (3,5]$ is disconnected subset of $\mathbb{R}$.

Reason:

Since there exists a pair of nonempty open subsets

$$U = (0,1) \quad \text{and} \quad V = (3,6)$$

of $\mathbb{R}$ such that

$A \cap U$ and $A \cap V$ are nonempty disjoint sets and

$$A = (A \cap U) \cup (A \cap V).$$

Activate Windows
Go to PC settings to activate Windows.

Examples: Connected Spaces

Example 2: Consider $X = \{a, b, c, d, e\}$ with $\mathcal{T} = \{\emptyset, \{a\}, \{a, b\}, \{a, b, c\}, \{a, d, e\}, \{a, b, d, e\}, X\}$.
Then $A = \{c, e\}$ is disconnected subset of X.

Reason:

Since there exists a pair of nonempty open subsets $U = \{a, b, d, e\}$ and $V = \{a, b, c\}$ of X such that

$$A \cap U = \{e\} \quad \text{and} \quad A \cap V = \{c\}$$

are nonempty disjoint sets and $\{e\} \cap \{c\} = \emptyset$

$$A = (A \cap U) \cup (A \cap V) = \{e\} \cup \{c\} = \{c, e\}$$

Activate Windows
Go to PC settings to activate Windows.

Examples: Connected Spaces

Example 3: Consider $X = \{a, b, c, d, e\}$ with $\mathcal{T} = \{\emptyset, \{a, b\}, \{a, b, c\}, X\}$. Then X is a connected space.

Reason:

Since there exists no pair of nonempty open disjoint subsets

$$U \text{ and } V$$

of X such that

$$X = U \cup V.$$

Activate Windows
Go to PC settings to activate Windows.

Examples: Connected Spaces

Example 4: Consider $X = \{a, b, c, d, e\}$ with $\mathcal{T} = \{\emptyset, \{a, b\}, \{d, e\}, \{a, b, c\}, \{a, b, d, e\}, X\}$. Then X is a disconnected space.

Reason:

Since there exist nonempty open disjoint subsets

$$U = \{a, b, c\} \text{ and } V = \{d, e\}$$

of X such that

$$X = U \cup V.$$

Activate Windows
Go to PC settings to activate Windows.

Connected Spaces and Open...



Recall

Connected Space:

A topological space (X, T) is said to be connected iff

$$X = D_1 \cup D_2$$

such that D_1, D_2 both open and disjoint implies that either

$$D_1 = \emptyset \quad \text{or} \quad D_2 = \emptyset.$$

Activate Windows
Go to PC settings to activate Windows.

Connected Spaces and Open...



Disconnected Space:
A topological space $(X, \mathcal{T})$ is said to be disconnected iff

$$X = D_1 \cup D_2$$

such that D_1, D_2 both open and disjoint implies that neither D_1 is empty nor D_2 is empty.

Activate Windows
Go to PC settings to activate Windows.

Connected Spaces and Open...

Consider

$$X = \{a, b, c, d, e\}$$

with

$$\mathcal{T} = \{\emptyset, \{a, b\}, \{d, e\}, \{a, b, c\}, \{a, b, d, e\}, X\}$$

$\{d,e\} = \{a,b,c\}$

and with

$$\mathcal{T}' = \{\emptyset, \{a, b\}, \{a, b, c\}, X\}.$$

X w.r.t $\mathcal{T}$ is disconnected.

Activate Windows
Go to PC settings to activate Windows.

Connected Set and its Closure



Theorem:
Let A be connected subset of a topological space (X, T) and

$$A \subset B \subset \overline{A}.$$

Then B is connected.

In particular, Closure of a connected subset is also connected.

Activate Windows
Go to PC settings to activate Windows.

Image of Connected Space



Theorem:
Let f be a continuous function from $(X, \mathcal{T})$ to $(Y, \mathcal{T}')$ and A be a connected subset of $(X, \mathcal{T})$. Then $f(A)$ is connected.
i.e.

Image of a connected space under a continuous map is connected.

Activate Windows
Go to PC settings to activate Windows.

Application: Image of...



Exercise I:
Consider $(\mathbb{R}, \mathcal{T}_u)$ and show that there exists no homeomorphism between $(0, 2)$ and $(0, 1] \cup [3, 4)$.

Exercise II:
Prove or disprove

$$(0, 1) \approx (0, 1]$$

Activate Windows
Go to PC settings to activate Windows.

Topology



Application: Image of Connected Set

Activate Windows
Go to PC settings to activate Windows.

Application: Image of...



Exercise I:
Consider $(\mathbb{R}, \mathcal{T}_u)$ and show that there exists no homeomorphism between $(0, 2)$ and $(0, 1] \cup [3, 4)$.

Exercise II:
Prove or disprove

$$(0, 1) \approx (0, 1]$$

Activate Windows
Go to PC settings to activate Windows.

Connected Subsets of Real Line



Theorem:
A subset A of $(\mathbb{R}, \mathcal{T}_u)$ is connected if and only if A is an interval.

Proof:
Let A be a nonempty connected subset of $\mathbb{R}$.
If A contains only one point then we have nothing to prove.
Let A contains two points, say

$$a, b \in A$$

Activate Windows
Go to PC settings to activate Windows.

Connected Subsets of Real Line

Consider $a < b$.
If every real number c, between a and b, belongs to A.
Then A is an interval. Proved.
If this is not the case, i.e. there exists some $c \in (a, b)$ such that

$$c \notin A$$

Then we can write A as

$$A = (A \cap (-\infty, c)) \cup (A \cap (c, \infty))$$

This provides us a separation of A, i.e. a nonempty, open, disjoint pair of sets whose union is A.

Activate Windows
Go to PC settings to activate Windows.

Connected Subsets of Real Line

This implies that A is disconnected and it is contradiction of the fact that A is connected.
So for all $c \in (a, b)$, $\implies$ $c \in A \implies$ A is an interval.
Converse:
Let A be an interval. i.e.

$$A = [a, b]$$

Now let A is disconnected. This means there exists a separation of A, i.e. there exist B, C nonempty, open, disjoint subsets such that

$$A = B \cup C \ .$$

Activate Windows
Go to PC settings to activate Windows.

Connected Subsets of Real Line

Now let

$$d = \sup\{c \in A | [a, c] \subset B\} \in A$$

Now one thing is very much clear here that $d \neq b$.
Let $d \in B$, now since B is open so there exists an open set U_d containing d such that

$$U_d \subset B.$$

So we can find a real number $d_1 \in U_d$ greater than d such that

$$[a, d_1] \subset B$$

Activate Windows
Go to PC settings to activate Windows.

Connected Subsets of Real Line

But this is a contradiction of the fact the d is the supremum. So $d \notin B$ and it means

$$d \in C$$

Again C is open so there exists an open set V_d containing d such that $V_d \subset C$.
This implies that there exists a real number $d_2 \in V_d$, less than d. So

$$[a, d_2] \subset B \quad \Rightarrow \quad d_2 \in B \cdot$$

And we have again a contradiction of the fact $B \cap C = \emptyset$.
So A is connected.

Activate Windows
Go to PC settings to activate Windows.

Connectedness and Fixed Point...



Fixed Point:

A point $x \in X$ is said to be a fixed point of a map $f : X \to X$ iff $f(x) = x$.

Example:

1. $f : \mathbb{R} \to \mathbb{R}$, defined as $f(x) = x^2$. Here 0 is a fixed point of f. 1 is also a fixed point of f.
2. $g : \mathbb{R} \to \mathbb{R}$, defined as

$$g(x) = x + 1$$

g has no fixed point.

Activate Windows
Go to PC settings to activate Windows.

Connectedness and Fixed Point...



Theorem:
Consider $\mathbb{R}$ with usual topology and

$$I = [0, 1]$$

be closed interval in $\mathbb{R}$. Then every continuous map

$$f : I \to I$$

has at least one fixed point.

Activate Windows
Go to PC settings to activate Windows.

Union of Connected Subsets



Theorem:
Let A and B be two connected subsets of a topological spaces (X, T) such that

$$x \in A \cap B$$

than

$$C = A \cup B$$

is a connected subset of X.

Activate Windows
Go to PC settings to activate Windows.

Union of Connected Subsets



Theorem:
The union of a collection of connected subspaces of $(X, \mathcal{T})$, that having a point common, is connected.

Activate Windows
Go to PC settings to activate Windows.

Connected Component



Def:
Consider a topological space $(X, \mathcal{T})$. Let $a \in X$ and C be a connected subset of X containing a.
Then

$$C_a = \bigcup_{a \in C} C$$

is called the connected component of X containing a.

Activate Windows
Go to PC settings to activate Windows.

Connected Component

④ A space $(X, \mathfrak{I})$ is connected iff it has only one connected component.

Remarks:

1. The connected component C_a is the largest connected subset of X containing a.
2. If $b \in C_a$ then $C_b = C_a$.
3. X can be written as disjoint union of its connected components. i.e.

$$X = \coprod_{C_i} C_i$$

Activate Windows
Go to PC settings to activate Windows.

Examples: Connected Component

Example 1: Consider $X = \{a, b, c, d, e\}$ with

$$\mathcal{T} = \{\emptyset, \{a, b\}, \{d, e\}, \{a, b, c\}, \{a, b, d, e\}, X\}$$

$C_a = \bigcup_{a \in C} C$

$= \{a, b, c\}$

$\{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}$

$\{a, b, c, d\} = (B \cap \{a, b, c\}) \cup (B \cap \{d, e\})$

B

$\Rightarrow C_a = C_b = C_c$

Now $C_d = \bigcup_{d \in C} C$

$= \{d, e\} = C_e$

$\{d\}$ $\{d, e\}$ $\{c, d, e\}$ Not Connected

$X = C_a \amalg C_d.$

Activate Windows
Go to PC settings to activate Windows.

Connected Component is Closed



Theorem:
Let C be a connected component of X then

$$C = \overline{C}$$

i.e.
A Connected component is a closed subset of X.

Activate Windows
Go to PC settings to activate Windows.

Connected Component is not Open



We have discussed about the closed nature of connected component of a topological space $(X, \mathcal{T})$.

Now one may ask the following question.

Question:

Is a connected component C of X open?

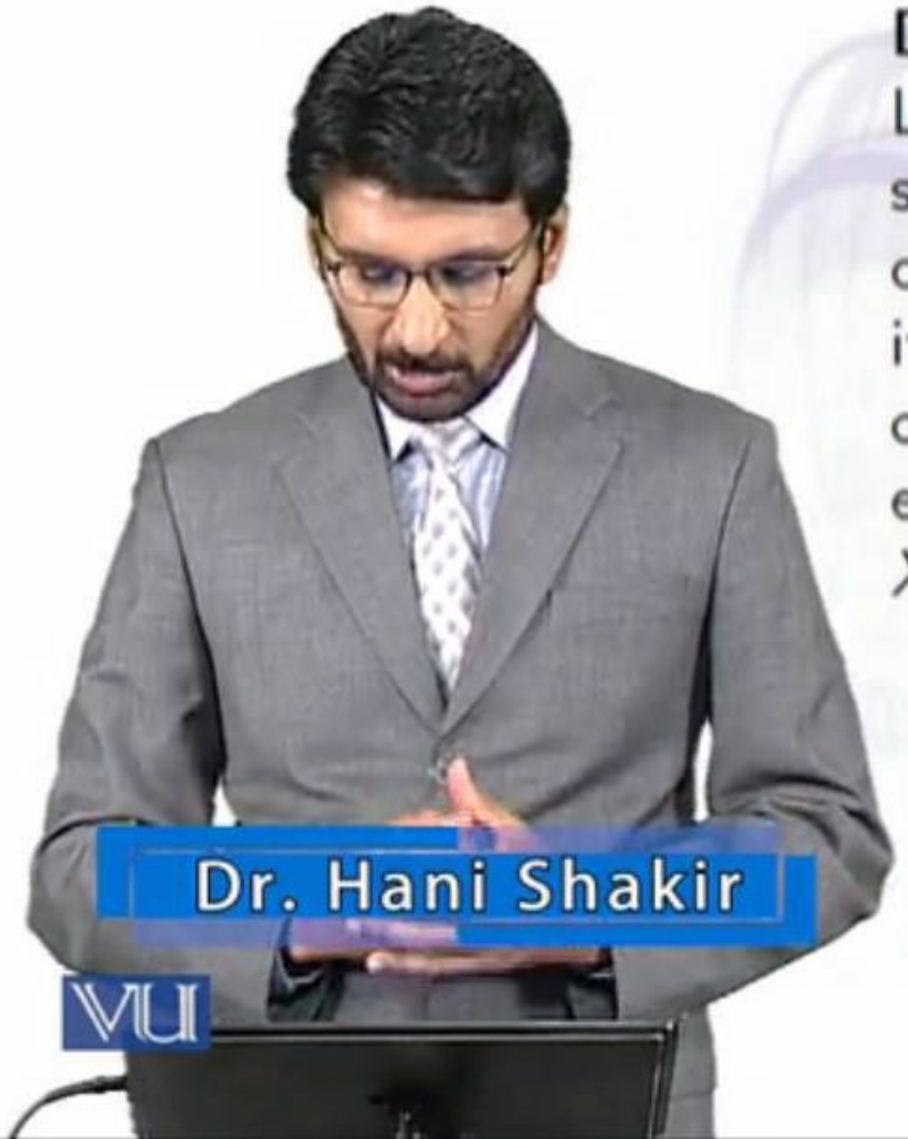
Answer:

No, not in general.

Lightshot
Screenshot is saved to Screenshot_129.jpg. Click here to open in the folder.

Activate Windows
Go to PC settings to activate Windows.

Locally Connected Spaces



Def:
Let $(X, \mathcal{T})$ be a topological space. X is said to be locally connected at a point $x \in X$ iff for every U_x, open subset of X containing x, there exists V_x, an open subset of X containing x, such that

- $V_x \subset U_x$
- V_x is connected.

Activate Windows
Go to PC settings to activate Windows.

Locally Connected Spaces



Def:

A topological space (X, T) locally connected iff it is locally connected at each of its points.

i.e.

For all $x \in X$ and for every U_x, open subset of X containing x, there exists V_x, an open subset of X containing x, such that

- $V_x \subset U_x$
- V_x is connected.

Activate Windows
Go to PC settings to activate Windows.

Examples: Locally Connected...

Example 2: A discrete space is locally connected.

X $\quad \tau_{dis} = \mathcal{P}(X)$

$x \in X$ $\quad U_x \in \mathcal{P}(X)$

$V_x = \{x\} \subset U_x.$

Activate Windows
Go to PC settings to activate Windows.

Connectedness and Locally...

Locally Connectedness $\not\Longrightarrow$ Connectedness

Example:

$$X = B(\underset{(-2,0)}{-2}; 1) \cup B(\underset{(2,0)}{2}; 1) \subset \mathbb{R}^2$$

X is not connected.

X is Locally connected.



Activate Windows
Go to PC settings to activate Windows.

Connectedness and Locally...

Connectedness $\not\Longrightarrow$ locally Connectedness

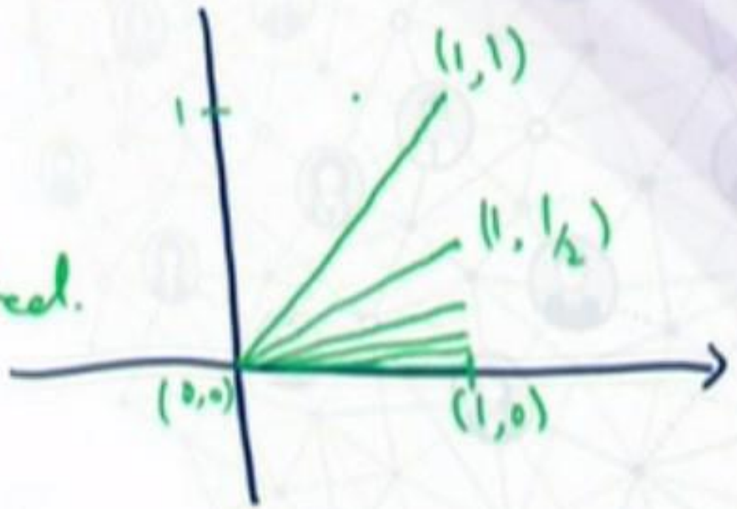
Example:

$$Y = \{(x, 0) | x \in [0, 1]\} \bigcup \left[\bigcup_{n \in \mathbb{Z}^+} \left\{ (x, \frac{x}{n}) | x \in [0, 1] \right\} \right] \subset \mathbb{R}^2$$

$n=1$ $\{(x, x) \mid x \in [0,1]\}$

$n=2$ $\{(x, \frac{x}{2}) \mid x \in [0,1]\}$

Now Y is connected.

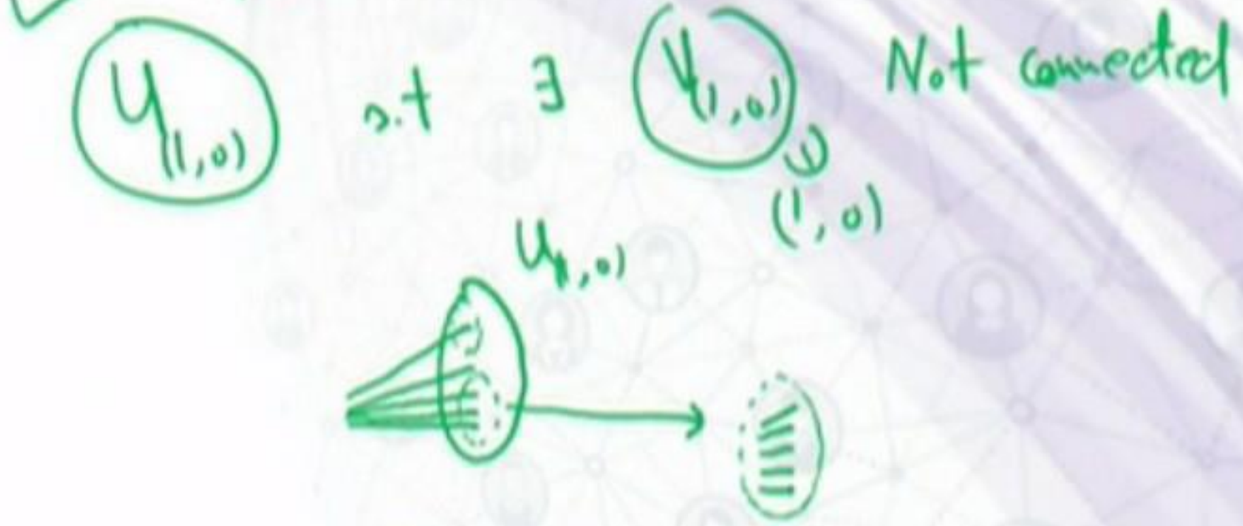


Activate Windows
Go to PC settings to activate Windows.

Connectedness and Locally...

Now Y is not Locally Connected.

$(1,0) \in Y$ s.t Y is not L.C at $(1,0)$



Activate Windows
Go to PC settings to activate Windows.

Path Connected Spaces



Path:
Let $(X, \mathcal{T})$ be a topological space and $x, y \in X$. A path from x to y in X is a continuous map

$$p : [0, 1] \to X$$

such that

$$p(0) = x$$

and

$$p(1) = y.$$

Activate Windows
Go to PC settings to activate Windows.

Path Connected Spaces



Path Connected Space:
A topological space $(X, \mathcal{T})$ is said to be path connected iff for every $x, y \in X$ there exists a path p from x to y in X.

Examples:

- $(\mathbb{R}, \mathcal{T}_u)$
- $(\mathbb{R}^2, \mathcal{T}_u)$

Activate Windows
Go to PC settings to activate Windows.

Facts: Path Connected Spaces: 1



Theorem:
Let $(X, \mathcal{T})$ be a path connected topological space then $(X, \mathcal{T})$ is connected.

i.e.

Path connectedness implies connectedness.

MTH 634

Module 133 To 183

Created by Badar Toor

MSc Math Bahawalnagar

fb.com/badartoor

Activate Windows
Go to PC settings to activate Windows.

Facts: Path Connected Spaces: 2



Proposition:
Let $(X, \mathcal{T})$ be a path connected topological space and

$$f : (X, \mathcal{T}) \longrightarrow (Y, \mathcal{T}')$$

be a continuous map. Then $f(X)$ is path connected.

Activate Windows
Go to PC settings to activate Windows.